

Bootstrapping

Covered by chapter 8.1, 8.3-8.4 in the textbook.

Typical ingredients in a statistical analysis of data are:

- Some observed data $x_1, x_2, \dots, x_n$.
- An assumed model for the distribution of the data.
- Use the data to find an estimate $\hat{\theta}$ for some parameter of interest θ .
- Calculate a confidence interval or make hypothesis test for θ .

What should we do if the data are clearly not normally distributed?

And what if we want to analyse other quantities than the mean or standard deviation (like the median, the rate, the quantile, the ratio, the correlation, ...)?

Example: Wave heights

A common model for wave heights is the Rayleigh distribution (which is a special case of the Weibull distribution) for which the pdf is

$$f(x) = \frac{x}{\theta^2} e^{-x^2/(2\theta^2)} \quad , x > 0, \quad \theta > 0.$$

To be able to use this distribution to e.g. calculate the probability of wave heights exceeding a certain height we need to have a (good) estimate of θ .

20 measurements of independent wave heights gave:

3.5	2.0	3.3	1.8	1.6	1.5	2.0	0.8	2.2	1.1
1.8	1.1	1.9	4.0	2.5	1.9	1.5	3.5	1.2	5.3

How to estimate θ from these data?

How to check uncertainty and bias of the estimate?

How to make a confidence interval for θ ?

The general bootstrap idea

The “true world”:

$$F \longrightarrow X_1, \dots, X_n \longrightarrow \hat{\theta}$$

The “bootstrap world”:

$$\hat{F} \longrightarrow X_1^*, \dots, X_n^* \longrightarrow \hat{\theta}^*$$

In the “true world”, **if** we knew F we could calculate/simulate anything of interest around $\hat{\theta}$, e.g. bias, standard deviation, confidence interval, distribution.

However, we do not know F . The bootstrap idea is to replace F by the estimate $\hat{F}$ (obtained from the data $X_1, \dots, X_n$) and then do all the calculations/simulations in this “bootstrap world” where we know $\hat{F}$.

If $\hat{F}$ is a good estimate of F , then calculations for $\hat{\theta}^*$ should be a good estimate for the same calculations for $\hat{\theta}$.

If the cdf in the “bootstrap world”, $\hat{F}$ is estimated by a parametric model we say that we do a *parametric bootstrap*.

If the cdf in the “bootstrap world”, $\hat{F}$ is estimated by the empirical cdf of the data we say that we do a *non-parametric bootstrap*.

Recall from chap. 2 that the ecdf is

$$\hat{F}(x) = \begin{cases} 0 & , x < x_{(1)} \\ \frac{i}{n} & , x_{(i)} \leq x < x_{(i+1)} \\ 1 & , x_{(n)} \leq x \end{cases}$$

where $x_{(1)} \leq x_{(2)} \leq \dots \leq x_{(n)}$ is the ordered data. I.e. $\hat{F}(x)$ jumps $\frac{1}{n}$ at each x_i .

Notice: To simulate from this distribution we just draw numbers uniformly from the observed data $x_1, \dots, x_n$.

The parametric bootstrap is a bit more statistically efficient than the non-parametric bootstrap *if* the assumed parametric model is correct. However, the non-parametric bootstrap is more robust as no (potentially wrong) distribution assumption is made. We shall only consider the non-parametric bootstrap and just refer to this as the bootstrap from now on.

Bootstrap resampling

To estimate the distribution of $\hat{\theta}$ (from which we can estimate bias, standard deviation and confidence intervals) we invoke the following algorithm:

1. Repeat step 2-3 for $b = 1, 2, \dots, B$.
2. Simulate data $x_1^{*(b)}, \dots, x_n^{*(b)}$ from $\hat{F}$ by drawing n observations *with replacement* from the original data $x_1, \dots, x_n$.
3. Calculate $\hat{\theta}^{*(b)}$ from $x_1^{*(b)}, \dots, x_n^{*(b)}$.

The ecdf of $\hat{\theta}^{*(1)}, \dots, \hat{\theta}^{*(B)}$ will be an estimate of the distribution of $\hat{\theta}$.

Example, Rayleigh:

Recall the wave data example where we based on measurements want to estimate the parameter θ in the pdf

$$f(x) = \frac{x}{\theta^2} e^{-x^2/(2\theta^2)} \quad , x > 0, \quad \theta > 0.$$

Solution: We first need to find a proper estimator for the parameter θ . We have learned earlier that this can be done by Maximum Likelihood Estimation (MLE).

Let $X_1, \dots, X_n$ i.i.d. from the Rayleigh distribution with pdf $f(x)$.

$$\begin{aligned} L(\theta) &= \prod_{i=1}^n f(x_i; \theta) = \prod_{i=1}^n \left(\frac{x_i}{\theta^2} e^{-x_i^2/(2\theta^2)} \right) \\ &= \frac{(\prod_{i=1}^n x_i)}{\theta^{2n}} e^{-\sum_{i=1}^n x_i^2/(2\theta^2)} \\ l(\theta) &= \ln L(\theta) = \ln\left(\prod_{i=1}^n x_i\right) - \ln(\theta^{2n}) - \ln(e^{-\sum_{i=1}^n x_i^2/(2\theta^2)}) \\ &= \ln\left(\prod_{i=1}^n x_i\right) - 2n \ln(\theta) - \frac{1}{2\theta^2} \sum_{i=1}^n x_i^2 \end{aligned}$$

$$l'(\theta) = -\frac{2n}{\theta} + \frac{1}{\theta^3} \sum_{i=1}^n x_i^2 = 0$$

$$2n\theta^2 = \sum_{i=1}^n x_i^2$$

$$\hat{\theta} = \sqrt{\frac{1}{2n} \sum_{i=1}^n X_i^2}$$

Check: $l''(\hat{\theta}) = -\frac{8n^2}{\sum_{i=1}^n x_i^2} < 0$ i.e. maximum.

Thus, the MLE approach gives us the estimator

$$\hat{\theta}_1 = \sqrt{\frac{1}{2n} \sum_{i=1}^n X_i^2}$$

From the data this gives the estimate $\hat{\theta}_1 = 1.76$.

Uncertainty? Bias? Mean square error? All these quantities can be computed/estimated using the bootstrap method:

$$\widehat{\text{bias}}(\hat{\theta}) = \overline{\hat{\theta}^*} - \hat{\theta}$$

$$\widehat{\text{SD}}(\hat{\theta}) = \widehat{\text{SD}}(\hat{\theta}^*) = \sqrt{\frac{1}{B-1} \sum_{b=1}^B (\hat{\theta}^{*(b)} - \overline{\hat{\theta}^*})^2}$$

$$\text{where } \overline{\hat{\theta}^*} = \sum_{b=1}^B \hat{\theta}^{*(b)} / B.$$

$$\text{MSE} = \text{E} \left[(\hat{\theta} - \theta)^2 \right] = \text{Var}(\hat{\theta}) + \left[\widehat{\text{bias}}(\hat{\theta}) \right]^2$$

See R code: `bootstrap_examples.R`

Bootstrap confidence intervals

To further reflect the uncertainty in the estimates we typically wish to calculate confidence intervals (CIs).

There are several ways the bootstrap approach can be used to obtain approximate CIs.

Standard normal bootstrap interval

If $\hat{\theta}$ is approximately normally distributed and unbiased then

$$Z = \frac{\hat{\theta} - E(\hat{\theta})}{SD(\hat{\theta})} = \frac{\hat{\theta} - \theta}{SD(\hat{\theta})} \sim N(0, 1)$$

and a $(1 - \alpha)100\%$ confidence interval is obtained by $[\hat{\theta} \pm z_{\alpha/2} \cdot SD(\hat{\theta})]$. Generally $SD(\hat{\theta})$ is unknown, but can be estimated by the bootstrap standard deviation estimate to obtain the interval:

$$\left[\hat{\theta} \pm z_{\alpha/2} \cdot \widehat{SD}(\hat{\theta}^*) \right]$$

Simple and easy to compute, but often not the best interval. Assumes approximate normality.

Percentile bootstrap interval

Let $\hat{\theta}^{*(1)}, \dots, \hat{\theta}^{*(B)}$ be the bootstrap estimates. The *percentile interval* is obtained by

$$\left[\hat{\theta}_{(\alpha/2)}^*, \hat{\theta}_{(1-\alpha/2)}^* \right],$$

where $\hat{\theta}_{(p)}^*$ denotes the p -th empirical quantile of bootstrap estimates.

No normality assumption. Naturally adapts to skewed bootstrap distributions. Easy to compute and interpret.

Basic bootstrap interval

The underlying idea is to approximate the distribution of $\hat{\theta} - \theta$ by the bootstrap distribution of $\hat{\theta}^* - \hat{\theta}$.

The *basic bootstrap interval* is obtained by reflecting the bootstrap quantiles around the observed estimate:

$$\left[2\hat{\theta} - \hat{\theta}_{(1-\alpha/2)}^*, 2\hat{\theta} - \hat{\theta}_{(\alpha/2)}^* \right].$$

BCa bootstrap interval

The bootstrap confidence intervals considered so far are easily implemented basic bootstrap confidence intervals. There exist several more sophisticated intervals. One of these is a modification of the percentile interval which includes a bias correction and an acceleration factor to speed up the convergence - called the BCa interval. See further details in the book, chap 8.4.

The BCa interval is less easy to implement, but there are routines in e.g. the `boot` library in R which can be used to implement the interval.

See R code: `bootstrap_examples.R`

How large B should be used?

For estimating confidence intervals it is recommended to simulate at least $B = 2000$ bootstrap samples. For estimating standard deviations and bias smaller B can be sufficient - but usually the simulation is so fast that we can use $B = 2000$ or more for these purposes as well.

Which interval to use?

Unfortunately there is no universally best bootstrap confidence interval. Empirical studies tend to give different recommendations in different situations. However, among those intervals discussed here the BCa is probably generally the best interval (and the standard normal bootstrap interval the least reliable).

Comparison of estimators:

The MLE approach gave us the estimator

$$\hat{\theta}_1 = \sqrt{\frac{1}{2n} \sum_{i=1}^n X_i^2}$$

Since in the Rayleigh distribution $E(X) = \theta\sqrt{\pi/2}$ and the expectation can be estimated by the average of the data another estimator is

$$\hat{\theta}_2 = \overline{X}\sqrt{2/\pi}$$

The latter estimator is called the moment estimator.

The MLE approach is generally recommended over the moment approach. MLE estimators have nice theoretical properties, but in some cases it is possible to find better estimators.

Which estimator is best in this case? We can study this by simulation.

Example, Rayleigh, compare estimators:

For the wave data, the two estimators give respectively

$$\hat{\theta}_1 = 1.76 \text{ and } \hat{\theta}_2 = 1.78$$

To compare the performance of $\hat{\theta}_1$ and $\hat{\theta}_2$ we can simulate many data sets from the Rayleigh distribution (using e.g. the the inverse transform algorithm) with a known value of θ and compare properties like bias, standard deviation and MSE of the estimators.

See R code: `bootstrap_examples.R`

Example: Bootstrap uncertainty on for spherical data

Let $\mathbf{X}_1, \dots, \mathbf{X}_n \in S^2$ be directional observations. Using the geodesic distance

$$d(\mathbf{x}, \mathbf{m}) = \arccos(\mathbf{x}^T \mathbf{m}),$$

the intrinsic sample Fréchet mean is

$$\hat{\boldsymbol{\mu}}_F = \arg \min_{\mathbf{m} \in S^2} \frac{1}{n} \sum_{i=1}^n d^2(\mathbf{X}_i, \mathbf{m}).$$

The uncertainty of $\hat{\boldsymbol{\mu}}_F$ can be investigated by bootstrap:

1. Resample $\mathbf{X}_1^*, \dots, \mathbf{X}_n^*$ with replacement.
2. Calculate the bootstrap Fréchet mean $\hat{\boldsymbol{\mu}}_F^*$.
3. Repeat B times.

Since the estimates lie on S^2 , uncertainty can naturally be measured by the angular distance $D^* = d(\hat{\boldsymbol{\mu}}_F^*, \hat{\boldsymbol{\mu}}_F)$. The 0.95 empirical quantile of the bootstrap values D^* gives an approximate 95% angular uncertainty radius around $\hat{\boldsymbol{\mu}}_F$.

See: `bootstrap_examples.R`